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ORIGINAL INVESTIGATION

Personalizing Resistance Training Mitigates Neuromuscular and Perceived Fatigue: The Autoregulation Cluster Training Method

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Purpose: To compare predetermined and autoregulated resistance training sessions on velocity loss and perceived fatigue. ***Methods:*** Twenty-six resistance-trained men completed 3 sessions including the back-squat and bench-press exercises matched for load (75% of 1-repetition maximum), volume (24 repetitions), and total rest (240 s). Sessions were randomly performed as traditional set (TRA), 3 sets of 8 repetitions with 120-second interset rests; cluster interset-rest redistribution (IRR), 6 clusters of 4 repetitions with 48-second between-clusters rests; and autoregulation cluster training (ACT), a personalized combination of clusters, repetitions per cluster, and between-clusters rest regulated on a velocity loss threshold. The comparative effects were evaluated on velocity loss outputs measured with a linear encoder and perceived fatigue responses reported using a single-item scale. ***Results:*** IRR and ACT induced less velocity loss than TRA ($b = -2.09$, $P < .001$). ACT also mitigated velocity loss more than IRR ($b = -2.31$, $P < .001$). The back squat resulted in greater velocity loss compared to the bench press ($b = 1.83$, $P < .001$). Perceived fatigue responses mirrored the pattern observed for the velocity loss outputs (IRR and ACT vs TRA: $b = -0.64$, $P < .001$; ACT vs IRR: $b = -1.05$, $P < .001$; back squat vs bench press: $b = 0.46$, $P = .005$). ***Conclusions:*** IRR and ACT reduced neuromuscular and perceived fatigue, likely due to their cluster-set structures' embedding frequent windows of interset rest. However, the ACT was overall more effective, presumably given its personalized structure.

Keywords: perception, velocity-based training, autonomy, velocity loss, load monitoring

Resistance training induces many structural and functional benefits on the musculoskeletal system.[1] Traditionally, resistance training has been prescribed according to recommendations of prominent health and professional organizations[2] in the form of rigid, nonpersonalized training programs and constituent training variables. To illustrate, fixed structures of sets, repetitions, and rest intervals are prescribed alongside percentages of the maximal load a person can lift for a single repetition (1-repetition maximum [1RM]) in an exercise. Although this one-size-fits-all approach is promoted at group level,[2] it fails to account for the variability of day-to-day performance.[3] To explain, 1RM capabilities may be reduced on a random day due to fatigue, sleep deprivation, or inadequate dietary regimens.[4,5] Moreover, the number of repetitions that can be completed against a given 1RM percentage is individual, exercise, and set specific.[3,6] Therefore, prescribing resistance training based on rigid structures and 1RM percentages may lead to an overloading stimulus in some days and an underloading in others.

Acute responses and chronic adaptations to resistance training occur as a complex interaction between training and nontraining factors,[4] with acute (ie, within session) and residual (ie, between sessions) fatigue playing an adverse role for optimal training processes.[7,8] A summary of 6 systematic reviews and meta-analyses[9–14] of the resistance training effects on neuromuscular, morphological, and functional adaptations highlights a few actionable insights. First, low within-set fatigue is required to optimize muscular strength and power development.[11,13] Second, no differences were observed in muscle mass growth between training to failure compared to training not to failure when volume was equalized.[9,10] Similarly, no differences were found between traditional-set (TRA) structures and less fatiguing alternative-set structures when 1RM percentages and volume were matched, which further demonstrates that considerable within-set fatigue is unnecessary to induce hypertrophy.[11] Finally, fatigue and muscle damage attenuate high-threshold motor unit recruitment and motor skill learning, thus impairing training quality.[15] This compelling evidence suggests avoiding excessive fatigue accrual as an imperative condition for optimal training effects.

A strategy to mitigate fatigue within a resistance training session is the cluster-set training approach.[16] Cluster-set training consists in splitting up the total number of repetitions performed into clusters (eg, 4–16) of fewer repetitions (eg, 2–6) interspersed by brief rest intervals (eg, 10–60 s). In particular, cluster-set training mitigates the psychological, perceptual, and metabolic stress associated with fatigue thereby ensuring greater neuromuscular outputs compared with TRA training that lacks any rest within a set.[12,14,16] However, while mean positive effects of cluster-set training are widely documented, a more individualized approach could mitigate fatigue further. In fact, all cluster-training variants are designed like the TRA approach by using predetermined and fixed configurations of clusters, repetitions per cluster, and between-cluster rest intervals.[16] This routine fails to accommodate individual needs and, depending on how the clusters (eg, repetitions per cluster and between-cluster rest) are structured, could still induce considerable fatigue for some. In support of this rationale, a recent study demonstrated that 2 cluster-set training variants, albeit superior to the TRA approach in mitigating fatigue, failed to optimize training outcomes when considering a predefined target.[17] Briefly, the training target was a proxy of neuromuscular fatigue—the 20% velocity loss—calculated from the first repetition in each set.[18] Between ~10% and ~50% of the participants were unable to maintain mean velocity outputs below the 20% velocity loss target across 18 repetitions in the bench press and the back

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squat exercises performed against a 10RM load. Therefore, a method that personalizes cluster-set configurations needs to be developed. To this aim, we conceptualized the autoregulation cluster training (ACT) method which builds upon and complements the cluster-set training approach by accommodating changes to cluster-set structures in a dynamic manner during the ongoing training session, based upon a specific autoregulation target. A description of the ACT method, its operational elements, and its implementation in the current study, is provided in the "Methods" section.

The purpose of this study was to investigate the comparative effects between TRA, fixed cluster-set, and ACT methods on velocity loss and perceived fatigue during resistance training sessions consisting of the bench press and back squat exercises loaded with 75% of 1RM. In line with the current literature, we hypothesized that both cluster-set and ACT methods will result in lower velocity loss and perceived fatigue responses compared to the TRA method. Second, considering the flexible, dynamic, and personalized nature of the ACT method, we hypothesized it will be more effective to mitigate velocity loss and perceived fatigue and, if not at least equal, to the fixed cluster-set training structure.

Methods

Subjects

Twenty-six resistance-trained men volunteered to participate in the study (Table 1). To be included in the study, participants had to: be healthy, be between the ages of 18 and 40 years, have at least 1 year of resistance training experience, and be able to lift 1RM loads in the bench press and back squat exercises ≥1.2 and ≥1.6 of their body weight, respectively. Written informed consent was obtained after the participants received an oral explanation of the purpose and potential risks of the study. All procedures were conducted in accordance with the Helsinki Declaration and approved by the University of the West of Scotland Ethics Committee.

Design

A randomized cross-over design was used to investigate the effects of predetermined and autoregulated resistance training sessions on velocity loss and perception of fatigue. Subjects reported to the laboratory on 5 occasions within a 2-week period. In the first session, subjects completed load–velocity relationship assessments in the bench press and back squat exercises. In the second session, they were familiarized with the study procedures. In the following 3 sessions, subjects performed the bench press and back squat exercises using 3 different training structures. Resistance training sessions were performed with 48 to 72 hours of rest between sessions. Barbell velocity was collected during each repetition, and rate of fatigue (ROF) was reported upon completion of each exercise within a 5-minute period.

Table 1 Demographic Data

	Mean (SD); range
Age, y	26.7 (3.2); 18 to 35
Height, m	1.73 (0.11); 1.61 to 1.91
Weight, kg	76.6 (10.3); 64.1 to 91.1
Resistance training experience, y	4.2 (1.1); 3 to 8
Estimated 1RM bench press, kg	109.4 (12.3); 84 to 151
Relative 1RM bench press/body weight	1.43 (0.3); 1.2 to 1.8
Estimated 1RM back squat, kg	138.2 (26.5); 102 to 184
Relative 1RM back squat/body weight	1.81 (0.5); 1.1 to 2.5

Abbreviation: 1RM, 1-repetition maximum.

Methodology

Load–Velocity Relationship Assessment

In the first session, load–velocity relationships in the bench press and back squat exercises were assessed for each subject. First, subjects completed a 15-minute general warm-up consisting of 5 minutes of arm cranking (Ergoselect 400, ergoline GmbH) and 5 minutes of running at self-selected paces, followed by 5 minutes of dynamic mobilization exercises. Second, 3 barbell (Eleiko) warm-up sets with progressively heavier loads (30%–80% of estimated 1RM) were performed for each exercise. Then, the load–velocity relationships were assessed following the protocol described elsewhere.[19] Subjects were given a passive rest of 10 minutes between the 2 relationship assessments. Barbell velocity was recorded during the concentric phase of each repetition using a linear encoder sampling at 1000 Hz (Chronojump). The greatest mean propulsive velocity values were then used to model the load–velocity relationships[19] and determine the 1RMs, whereby we calculated the individual 75% 1RM loads to use in the resistance training sessions.

Resistance Training Sessions

Following the same warm-up protocol described above, subjects completed 3 resistance training sessions including the bench press and back squat exercises, matched for load (75% 1RM), volume (24 repetitions) and total rest (240 s), which were performed according to 3 structures:

1. TRA: Three sets of 8 repetitions with 2 minutes of between-set rest
2. Interset rest redistribution (IRR): Six clusters of 4 repetitions with 48 seconds of between-cluster rest
3. ACT: A personalized structure of clusters, repetitions, and rest intervals regulated according to a 20% velocity loss calculated from the first repetition of the training session. Specifically, the volume of 24 repetitions was split into clusters, with the number of repetitions per cluster determined upon the repetitions performed until reaching the autoregulation target at the first attempt (ie, commencement of each exercise). The overall rest duration of 240 seconds was evenly distributed between the number of clusters required to complete the volume. To illustrate, a trainee may complete 4 repetitions before reaching the 20% velocity loss target. Considering a volume of 24 repetitions and 240 seconds of rest, the ACT structure was: 1 cluster (ie, first cluster) of 4 repetitions and 7 clusters of 3–3–3–3–3–3–2 repetitions (20 remaining repetitions after the first cluster divided by 3 given that the last repetition in the first cluster fell above the 20% velocity loss target) with ~34 seconds (240 s divided by 7 between-cluster intervals) of between-cluster rest. The ACT method differed from the IRR given its flexibility to accommodate changes of exercise structures through an iterative process. Referring to the same example, in the second or any successive cluster, the trainee may have reached the autoregulation target before completing 3 repetitions as initially determined, thus carrying over one or more

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repetitions. Here, the ACT approach allowed recalculation of the number of clusters necessary to complete the number of repetitions left over. Accordingly, the remainder of the overall rest duration was split across the updated number of clusters. This iterative process was perpetuated until the 24 repetitions were completed. A video example of the ACT method in action is accessible through the following link: https://youtu.be/dRdLgLxo8EI.

The order of the 3 resistance training sessions and the 2 exercises within each session was randomized (www.random.org). A 10-minute period of passive recovery was provided between the 2 exercises during each session. Sessions were conducted by the same 2 researchers at approximately the same time of (2:00–6:00 PM). Verbal encouragement was provided to ensure that subjects performed the concentric phase of the exercises with the maximal velocity. During the resistance training sessions, if the velocity of the first repetition deviated (±0.05 $m \cdot s^{-1}$) from the value(s) observed during the other condition(s), subjects were instructed to terminate the ongoing set immediately. Researchers then promptly recalculated and adjusted the load to accommodate the initial velocity before the session could be resumed.[18] Across the total of 156 resistance training sessions completed in this study, this occurred 3 times among 3 different participants. Subjects were asked to refrain from intense training targeting muscle groups involved in the bench press and back squat exercises for at least 48 hours prior to the resistance training sessions, to avoid confounding effects due to muscular fatigue and soreness.

Bar Velocity Data Processing

Mean propulsive velocity outputs were collected using the same linear encoder described above, which was fixed to the barbell at a perpendicular angle to the floor during the bench press and back squat exercises. Data collection and processing followed the procedures described elsewhere[19] using the customary software provided by the manufacturer. Data were then exported into a Microsoft® Excel spreadsheet (Microsoft) to: (1) calculate velocity loss values (in percentages) in each session using the velocity of the first repetition of the training as reference (Table 2). Accordingly, velocity loss values were calculated from the 2nd to the 24th repetition and (2) explore individual cluster-set configurations in both exercises completed during the ACT condition. The summary data set of the individual ACT configurations per exercise is available in Supplementary Material S1 (available online).

Rate of Fatigue

The single-item, 11-point ROF scale was used to assess perceived nonlocal muscular fatigue.[20] The scale ranges from 0 ("not fatigued at all") to 10 ("total fatigue"). The question "How fatigued are you at the moment?" was presented at the top of the scale. Subjects were asked to report their ROF within 5 minutes upon completion of each exercise. The ROF data set is available in Supplementary Material S2 (available online).

Statistical Analysis

Descriptive data are presented as mean (SD) unless otherwise stated. To examine the effects of resistance training structure and exercise on velocity loss and perceived fatigue, we used the following linear mixed-effects model:

$$\text{Outcome}_{\text{in}} = b_{0\text{in}} + b_{1\text{–}3}\text{training structure} \times b_{4\text{–}5}\,\text{exercise}_{\text{in}} + \epsilon_i.$$

Table 2 Experimental-Sessions Data

	Mean (SD); range
Velocity first repetition, $m \cdot s^{-1}$	
Bench press	
TRA	0.57 (0.03); 0.53 to 0.61
IRR	0.57 (0.02); 0.54 to 0.60
ACT	0.58 (0.03); 0.53 to 0.62
Back squat	
TRA	0.74 (0.05); 0.68 to 0.80
IRR	0.75 (0.06); 0.68 to 0.82
ACT	0.74 (0.05); 0.68 to 0.80
Velocity loss, %	
Bench press	
TRA	19.8 (13.0); –6.1 to 68.1
IRR	15.9 (11.6); –7.0 to 68.8
ACT	11.3 (8.0); –9.6 to 43.2
Back squat	
TRA	23.3 (14.5); –6.1 to 69.0
IRR	16.6 (11.8); –7.7 to 71.2
ACT	12.6 (8.8); –7.7 to 47.9
Rate of fatigue, AU	
Bench press	
TRA	6.5 (1.4); 3 to 9
IRR	5.6 (0.9); 4 to 7
ACT	3.5 (1.1); 2 to 7
Back squat	
TRA	7.1 (1.3); 4 to 10
IRR	5.8 (1.3); 3 to 8
ACT	4.2 (1.0); 2 to 6

Abbreviations: ACT, autoregulation cluster training; AU, arbitrary unit; IRR, interset rest redistribution; TRA, traditional.

Velocity loss (in percentages) or ROF (arbitrary unit [AU]) represented the repeated-measures *outcome* for subject$_{in}$ and served as outcome measures, whereas *training structure* (categorical variable with 3 levels [TRA, IRR, and ACT]) and *exercise* (categorical variable with 2 levels [bench press and back squat]) were modeled as predictors and treated as fixed effects. Two contrasts were set on the variable *training structure* to examine the 2 hypotheses of this study: (1) effectiveness of IRR and ACT methods over the TRA method and (2) effectiveness of the ACT method over the IRR method. Interactions between *training structure* and *exercise* were also included in the model. Moreover, random effects were assumed for participants, with random slopes introduced in the model if their addition did not result in a convergence error. Estimated marginal means and 95% confidence intervals (CIs) were calculated alongside comparisons made using post hoc Holm–Bonferroni adjustments. Visual inspection of residual plots was used to confirm the assumptions of homoscedasticity or normality, which was also assessed through the Shapiro–Wilk test. Moreover, since regression models can be sensitive to multicollinearity, we computed the variance inflation factors for all predictor parameters used in the linear mixed-effects model to inspect the presence of autocorrelation between pairs of predictors. All statistical analyses were conducted in R language and

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environment for statistical computing using the *lme4*, *lmerTest*, *emmeans*, and *ggeffects* packages, while model assumptions were checked using the *performance* package (4.0.5, R Core Team). The data analysis code is available here: https://osf.io/29yxp.

Results

For descriptive purposes, outputs of the resistance training sessions are presented as means, SD, and range in Table 2. Frequency distributions of the individual training structures consisting of number of clusters and repetitions across exercises during the ACT condition are shown in Figure 1. The raw individual data points for velocity loss and ROF grouped by training structure across the 2 exercises are shown in Figures 2 and 3, respectively. Results from the linear mixed-effects models are presented in Tables 3 and 4.

The linear mixed-effects model on ROF did not reveal significant interactions between training structure and exercise, neither when comparing IRR and ACT pooled against TRA nor when IRR was compared to ACT ($P = .136$; $b = -0.81$; 95% CI, −1.33 to 1.05; percent loss) nor when IRR was compared to ACT ($P = .464$; $b = 0.34$; 95% CI, −0.57 to 1.24; percent loss). However, a main effect was observed for training structure (TRA > IRR > ACT), with contrasts revealing lower velocity loss for IRR and ACT pooled together compared to TRA ($P < .001$; $b = 2.09$; 95% CI, −2.46 to −1.72; percent loss) and lower velocity for ACT compared to IRR ($P < .001$; $b = -2.31$; 95% CI, −2.95 to −1.67; percent loss). In addition, a main effect was observed for exercise, with the back squat resulting in greater velocity loss compared to the bench press ($P < .001$; $b = 1.83$; 95% CI, 1.09 to 2.57; percent loss).

The linear mixed-effects model on ROF did not reveal significant interactions between training structure and exercise, either when comparing IRR and ACT pooled against TRA ($P = .711$; $b = -0.04$; 95% CI, −0.27 to 0.19 AU) or when IRR was compared to ACT ($P = .262$; $b = 0.23$; 95% CI, −0.17 to 0.62 AU). However, a main effect was observed for training structure (TRA > IRR > ACT), with contrasts revealing lower ROF for IRR and ACT pooled together compared to TRA ($P < .001$; $b = -0.64$; 95% CI, −0.81 to −0.47 AU) and lower ROF for ACT compared to IRR ($P < .001$; $b = -1.05$; 95% CI, −1.34 to −0.75 AU). In addition, a main effect was observed for exercise, with the back squat resulting in greater ROF compared to the bench press ($P = .005$; $b = 0.46$; 95% CI, 0.14 to 0.79 AU).

Discussion

In this study, we compared the effects of predetermined and autoregulated resistance training sessions, matched for load, volume, and rest on velocity loss and perceived fatigue. Subjects performed the bench press and back squat exercises as a TRA, IRR, and personalized ACT structure. Two main findings emerged. First, and aligned with our hypotheses, the IRR and ACT induced lower velocity loss in both exercises compared to the TRA. Moreover, the ACT was more effective in mitigating velocity loss compared to the IRR. Overall, greater velocity loss occurred in the back squat compared to the bench press exercise. Second, the perceived fatigue responses mirrored the pattern observed for the velocity loss outputs.

The lower velocity loss observed during the IRR and ACT compared to the TRA are in line with previous investigations

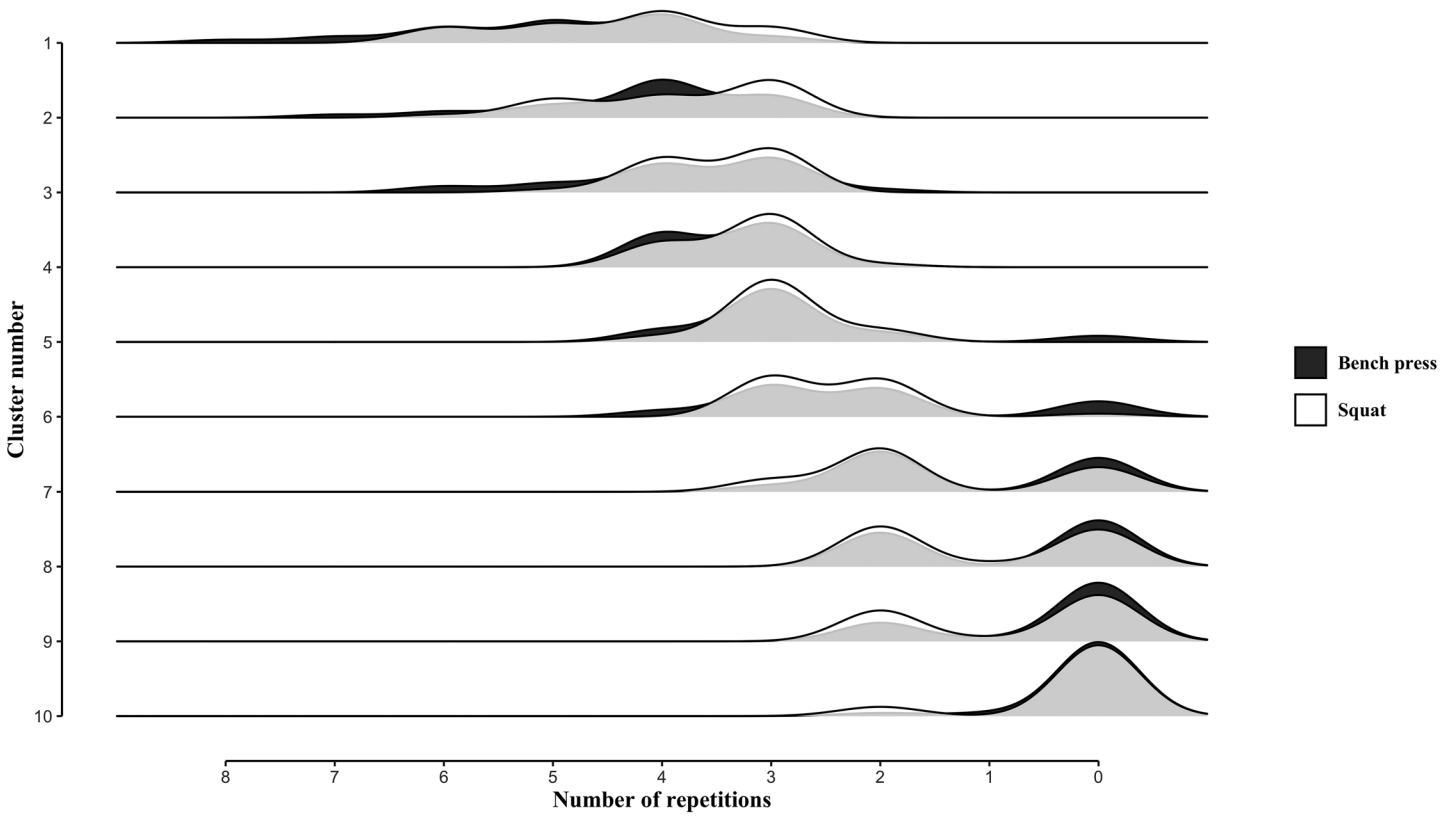

Figure 1 — Frequency distributions of the individual autoregulation cluster-training configurations grouped by exercise.

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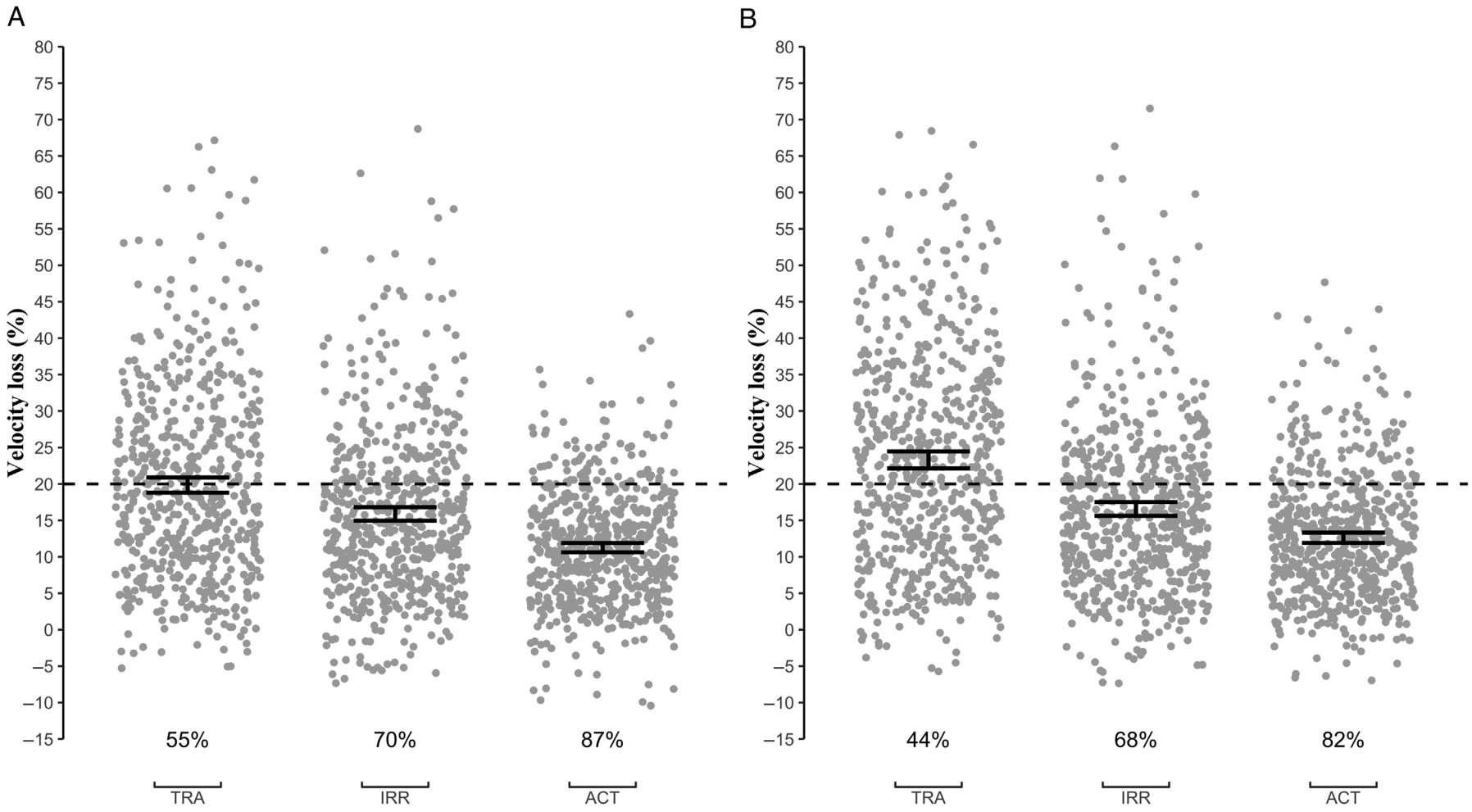

Figure 2 — Velocity loss outputs across training structures in the bench-press (panel A) and back-squat (panel B) exercises. The *y* intercept represents the 20% velocity loss target. Error bars display the 95% CI. Percentages represent the proportion of repetitions performed within the 20% velocity loss threshold. ACT indicates autoregulation cluster training; IRR, interset rest redistribution; TRA, traditional.

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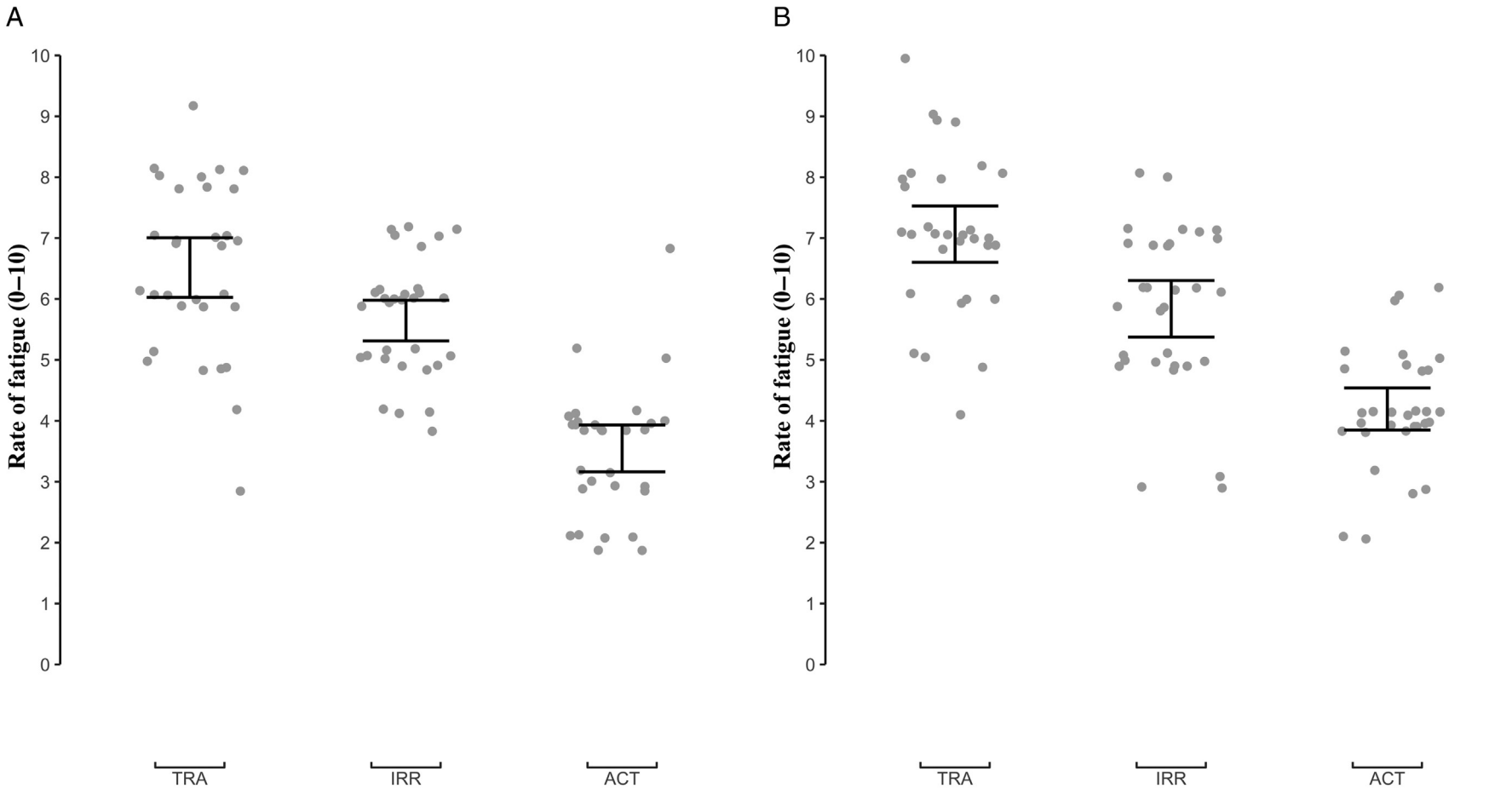

Figure 3 — Rate of fatigue outputs across training structures in the bench-press (panel A) and back-squat (panel B) exercises. Error bars display the 95% CI. ACT indicates autoregulation cluster training; IRR, interset rest redistribution; TRA, traditional.

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Table 3 Linear-Regression Fixed Effects (Within Subject) Outputs of the Velocity Loss

	Coefficient	SE
Constant	15.67	0.56
Traditional		
IRR and ACT	−2.09***	0.19
IRR vs ACT	−2.31***	0.33
Bench press		
Back squat	1.83***	0.38
IRR and ACT × back squat	−0.81	0.96
IRR vs ACT × back squat	0.34	0.46

Abbreviations: ACT, autoregulation cluster training; IRR, interset rest redistribution.
***$P < .001$.

Table 4 Linear-Regression Fixed Effects (Within Subject) Outputs of the Rate of Fatigue

	Coefficient	SE
Constant	5.24	0.12
Traditional		
IRR and ACT	−0.64***	0.09
IRR vs ACT	−1.05***	0.15
Bench press		
Back squat	0.46**	0.16
IRR and ACT × back squat	−0.04	0.12
IRR vs ACT × back squat	0.23	0.20

Abbreviations: ACT, autoregulation cluster training; IRR, interset rest redistribution.
**$P < .01$. ***$P < .001$.

promoting cluster-set training to mitigate acute neuromuscular fatigue.[12,16,19] We assume that the short and frequent windows of rest embedded within the IRR and ACT structures reduced the rate of adenosine triphosphate depletion and favored quicker regeneration of phosphocreatine within the working muscles thereby allowing maintenance of greater velocity outputs.[21] Moreover, the greater velocity loss during the TRA may have been caused by an acute increase in metabolic stress. Although speculative, our assumption is plausible as compelling evidence demonstrated that TRA structures induce greater blood lactate concentration and the consequent inability to avoid velocity loss across repetitions in the ongoing set and between consecutive sets.[21,22] Irrespective of the underpinning mechanisms, it is noteworthy that both IRR and ACT are efficient methods as they do not extend the total training time compared to the TRA.

The novel finding of this study is that the ACT method is effective to attenuate velocity loss further, even more than the IRR despite the latter also embedding a cluster-set structure. We assume that the flexible, dynamic, and personalized nature of the ACT method favors individual abilities. This assumption is supported by the large range and variety of individual combinations, including clusters, number of repetitions per clusters, and between-cluster rest intervals, we observed across both exercises in the ACT condition (Figure 1). By allowing people to adjust a personalized ACT structure during the ongoing session, rather than a predetermined structure, individual differences in abilities are better accounted for.[23,24] This is central in resistance training as the number of repetitions performed toward autoregulation targets, such as velocity loss thresholds, rates of effort, or proximity to task failure, are individual, exercise, and set specific.[3,6,25,26] Moreover, the ACT method seems particularly effective in combination with the velocity-based training approach which requires sets to be terminated upon exceeding velocity loss targets. In fact, the ACT method facilitates performing most of the repetitions within the velocity loss threshold. To illustrate, in the ACT condition the percentages of repetitions performed within the 20% velocity loss target was 87% and 82% in the bench press and back squat exercises, respectively (Figure 2). While future studies should confirm the effectiveness of the ACT across a broader range of velocity loss targets, this novel method seems promising for managing and mitigating neuromuscular fatigue in a personalized manner.

The velocity loss observed in the back squat was ~2 percentage points greater compared to the bench press. These results are in line with those of previous studies[27,28] but in contrast with others[17,29] comparing velocity loss between the same exercises performed with similar loads (60%–75% 1RM). Several methodological differences that could explain these contrasting results include: the velocity metric to calculate the velocity loss (mean velocity vs mean propulsive velocity), exercise variants (parallel squat vs full squat), the presence or absence of a pause between the eccentric and concentric phases, and the lifting velocity (controlled by means of a metronome vs maximal intended velocity). Yet, differences in velocity loss observed between the 2 exercises confirm that neuromuscular relationships and performance for upper and lower body muscles are exercise specific.[25,26]

As ROF responses reflected the same patterns of the velocity loss outputs, it appears that the 2 cluster-set structures were also perceived as less fatiguing with the ACT being least fatiguing. While the assumptions based upon the likely differences between training structures in energy substrate restoration and lactate concentration may hold true for the lower ROF responses following both cluster-set structures, we consider another reason to explain the differences between the IRR and ACT. The ACT method is advantageous as its cluster-set structure is individually tailored and can also be adjusted upon the performance outputs during the ongoing session whenever necessary. This allows for a more personalized representation of one's needs, thus presumably mediating greater perception of control and enhanced motivation to pursue performance goals.[23,30] In light of the above and in consideration that perception of fatigue is a key determinant in resistance training, it seems prudent that ACT could be advocated when prescribing training in a personalized manner.

This study has a few limitations. First, we only included male subjects, who performed 2 exercises. It is thus unclear if our findings generalize to other populations and exercises. Second, the resistance training sessions included only one load corresponding to 75% 1RM. Third, we only used one cluster-set variant as an alternative to TRA and ACT structures. It may well be the case that the effectiveness of the ACT method may vary in comparison to other cluster-set structures or configurations. Finally, given that biological markers of metabolic stress (eg, lactate concentration) were not collected, we are unable to elaborate further on the underlying mechanisms.

Practical Applications

Practitioners should consider using the ACT method as a strategy to prescribe resistance training in a personalized manner. The ACT

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method seems to exploit velocity outputs by mitigating neuromuscular and perceived fatigue. Utilizing the ACT method is an effective approach as it does not require any additional training time compared to other TRA approaches.

Conclusions

We observed lower velocity loss and perceived fatigue in the IRR and ACT compared to the TRA structure. These outcomes likely stem from rest intervals more frequently integrated between repetitions within the cluster-set structures which led to reduced neuromuscular and perceived fatigue. However, the ACT was overall more effective, presumably due to its personalized structure.

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